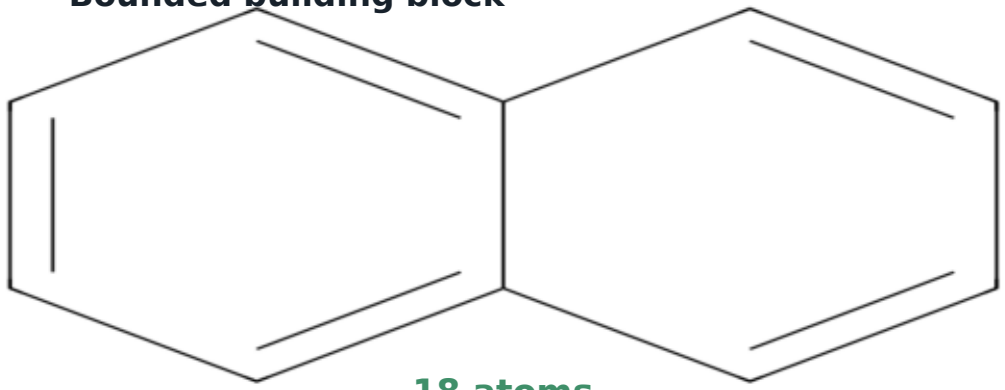
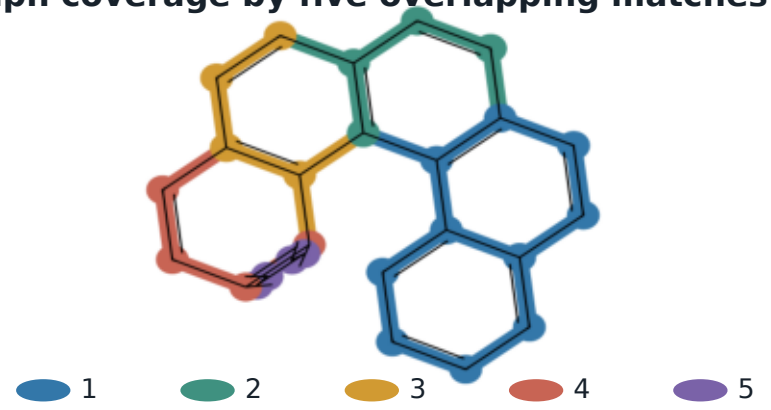
Overlap-constrained assembly: from an 18-atom tile to [6]helicene

A **Bounded building block**



18 atoms

B **Graph coverage by five overlapping matches**



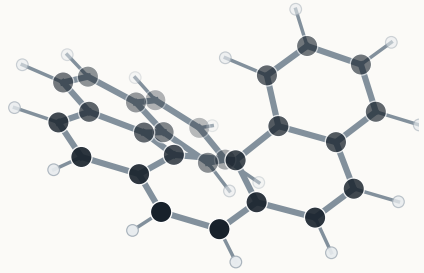
C **Overlap fixes the three-dimensional placement**



6 shared, non-collinear C atoms

mean overlap RMSD = 0.012 Å · no link-axis ambiguity

D **Steric release gives an Avogadro-ready helix**



[6]helicene · $C_{26}H_{16}$ · 42 atoms

fragment library
substructure matches

→

overlap assembly
bounded, auditable

→

ORACLE
perceive + freeze state

→

Avogadro
inspect / edit XYZ

→

applications
consume frozen state

The continuous perception record is the computational state passed downstream.